

Q: Dealing with outliers with rotation.

Consider the matrix R and given vector x
if $x_i > M$ / M = defined as some clipper for

for some i if $x_i > M \rightarrow \Pr(\text{clip } x_i) = 1$
usual setup

Now $\rightarrow y = R x$

- Random Gaussian Matrix

- Random projection Matrix

- Almost orthogonal.

(eigenvalues are $(1 \pm \epsilon)$)

Let us look at

$$y_i = \vec{r} \cdot \vec{x}$$

$$= \sum r_i x_i$$

$$r_i \approx \frac{1}{\sqrt{d}} \cdot N(0, 1)$$

$$E(y_i) = 0$$

linearity of expectation

$$\text{Var}(y_i) = \sum x_i^2 \cdot \text{Var}(r_i)$$

$$= \|x\|^2 \cdot \text{Var}\left(\frac{1}{\sqrt{d}} N(0, 1)\right)$$

$$\text{Var}(y_i) = \frac{\|x\|^2}{d}$$

$$y_i \sim N\left(0, \frac{\|x\|}{\sqrt{d}}\right)$$

$$\text{Prob}(|y_i| > M) = 2 \phi\left(-\frac{M\sqrt{d}}{\|x\|}\right)$$

CDF of normal dist.

Chernoff
bound

$$\text{Pr}(|y_i| > M) \leq e^{-\frac{M^2 d}{2 \|x\|^2}}$$

Thus the probability of clipping goes down exponentially in "d".